QP Code : 26900258

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) /MGU-BBA (Honours)/ MGU-BCA

(Honours)Regular EXAMINATION October 2026

Fifth Semester

Discipline Specific Core Course (DSC) - MG5DSCGEO300 -

Structural Geology

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Part A
Short Answer in 60 words
Answer any 7 questions
Each question carries 2 marks

1. Distinguish between true dip and apparent dip.- True dip is the maximum angle of inclination of a geological plane from the horizontal, measured perpendicular to its strike. Apparent dip is the angle measured in any direction other than perpendicular to strike; it is always equal to or less than the true dip. It becomes equal to true dip when measured perpendicular to strike and zero when measured parallel to strike.

2. Tectonite fabric - Tectonite fabric is the preferred arrangement of mineral grains, clasts, or structural elements produced by directed stress and deformation in a rock. It may be planar (foliation), linear (lineation), or a combination of both. Fabric records the orientation and style of deformation and is useful in interpreting tectonic history.

3. Hanging wall and footwall - In an inclined fault, the hanging wall is the rock mass above the fault plane, while the footwall is the rock mass below it. These terms are geometric and do not by themselves indicate the direction of movement. In a normal fault the hanging wall moves down relative to the footwall; in a reverse fault it moves up.

4. Parallel fold - A parallel fold is a fold in which the thickness of a bed, measured perpendicular to bedding, remains approximately constant around the fold. The beds are therefore approximately parallel to one another through the folded structure.Parallel folds are commonly described as concentric folds and contrast with similar folds, in which layer thickness measured perpendicular to the axial plane is approximately constant.

5. Stratigraphic hiatus. - A stratigraphic hiatus is a gap in the geological record representing a

period during which sediments were not deposited, or previously deposited material was eroded. It may be recognized as a break in succession and can be associated with an unconformity. Hiatuses are important because they indicate missing time in the rock record.

6. Plunge - Plunge is the angle that a linear geological feature such as a fold hinge, mineral lineation, or intersection line—makes with the horizontal, measured in a vertical plane. It is described together with a trend or bearing.

7. Effect of temperature on deformation.- Increasing temperature generally promotes ductile deformation by facilitating crystal plasticity, recrystallization, and diffusion. Rocks may deform at lower differential stress and show less brittle fracturing. At lower temperatures and shallow crustal levels, deformation is more commonly brittle. The response also depends on pressure, strain rate,mineral composition, fluids, and time.

8. Systematic joints.- Systematic joints are fractures that occur in organized sets with broadly consistent orientation and spacing.Individual joints in a set are commonly approximately parallel. They may form due to tectonic stress, cooling,unloading, or other processes. Two or more intersecting sets can divide rock into blocks and influence weathering,groundwater flow, and engineering stability.

Part B

Short notes in 250 words

Answer any 3 questions

Each question carries 8 marks

9. Explain strike and its geological significance.

Strike is the compass direction of the horizontal line contained in an inclined planar surface, such as a bedding plane, foliation, fault plane, or joint. It is usually expressed as a bearing or azimuth from 000° to 360°. Strike is at right angles to the true-dip direction. A horizontal bed has no unique strike, while a vertical plane has a strike but its dip direction can be ambiguous unless stated by convention or field observation.Measurement: In the field, a geological compass or compass-clinometer is placed against the planar surface and levelled to measure the horizontal line. The strike is recorded together with dip direction and dip angle using a consistent convention.**Geological significance**: (1) Strike helps map the distribution and continuity of beds and structures. (2) It is used with dip to determine the attitude of planar features. (3) Changes in strike can help identify folds, bends, and fault-related rotation. (4) Strike measurements support geological cross-sections and structural maps. (5) They help interpret regional structural trends and the orientation of joints, faults, cleavage, and foliation. (6) In engineering geology, strike and dip inform slope, tunnel, foundation, and excavation assessments.(Definition 2 marks, Significance 6 marks)

10. Compare planar fabric and linear fabric in deformed rocks.

Planar fabric is a preferred arrangement of mineral grains or structural elements in a plane. Examples include slaty cleavage, schistosity, gneissic banding, and compositional layering. It may develop through mineral alignment,flattening, pressure solution, segregation, or metamorphic differentiation. It is described by its orientation and continuity.Linear fabric is a preferred alignment of elongated or linear elements along a direction. Examples include mineral lineation, stretching lineation, and fold-hinge lineation. It may form through stretching, mineral growth, intersection of planar fabrics, or alignment of elongated clasts and minerals.Planar fabric is two-dimensional and is measured as a plane using strike and dip; linear fabric is one-dimensional and is measured by trend and plunge. Planar fabrics commonly record flattening or layer-parallel alignment, whereas linear fabrics may indicate stretching direction, mineral growth direction, or hinge orientation.Both can occur together in deformed rocks and provide complementary evidence of strain and tectonic transport.(Planar fabric - 4 marks,Linear Fabrics - 4 marks ,Explain each with figures)

11. Classify faults based on net slip. (Briefly explain with diagram)

Net slip is the total displacement of one fault block relative to the other, measured along the fault plane.Classification based on net-slip direction includes:

1. Dip-slip faults: Net slip is parallel to the dip direction of the fault plane. Normal faults form where the hanging wall moves down relative to the footwall and are commonly associated with extension. Reverse faults form where the hanging wall moves up relative to the footwall and are associated with shortening. A low-angle reverse fault is commonly called a thrust fault.

2. Strike-slip faults: Net slip is approximately parallel to strike and horizontal. In a right-lateral (dextral) fault, the opposite block appears to move to the right; in a left-lateral (sinistral) fault, it appears to move to the left.

3. Oblique-slip faults: Movement has both dip-slip and strike-slip components. They combine vertical and horizontal displacement and may reflect more complex stress conditions.The sense of net slip is determined from displaced markers, slickensides, steps on fault surfaces, drag structures,and other kinematic indicators.

12. List the main field indicators of fold.(Any 8 points - each point carry 1 marks)

- Folds are recognized in the field through several geometric and stratigraphic indicators:
- Repeated or inverted strata: Older and younger beds may repeat across an anticline or syncline; overturned limbs can reverse the normal stratigraphic order.
- Changes in strike and dip: Systematic changes in bedding attitude can define fold limbs and the hinge region.
- Curved bedding traces: Outcrop patterns and map traces may show arcuate or repeated bands, depending on topography and fold orientation.

- Closure of outcrop patterns: Closed or nose-shaped patterns may indicate plunging folds when interpreted with topography and younging directions.
- Axial-planar cleavage or foliation: A planar fabric may be approximately parallel to the axial plane and may become more closely spaced near hinges.
- Minor folds and parasitic folds: Small-scale folds can reveal the orientation and geometry of larger structures.
- Hinge zones and fold-axis lineations: Bedding curvature and lineations can help locate hinge lines and determine plunge.
- Younging indicators: Graded bedding, cross-bedding, ripple marks, and other way-up structures help distinguish upright from overturned limbs.

13. Differentiate the major types of stress and strain in rocks.

Stress is force per unit area acting within a rock, whereas strain is the change in shape or size produced by deformation.Types of stress: (1) **Compression** acts to shorten a rock and commonly produces folds, reverse faults, and thrusts. (2) **Tension or extension** acts to lengthen or pull apart a rock and may produce joints and normal faults. (3)**Shear stress** acts parallel to a surface and causes adjacent parts of a rock to slide past one another. **Lithostatic pressure** acts approximately equally in all directions and differs from differential stress.Types of strain: (1) **Elastic strain** is recoverable when stress is removed. (2) **Plastic or ductile strain** is permanent and may produce folds, cleavage, and mineral fabrics. (3) **Brittle strain** involves fracture and displacement, forming joints and faults. (4) **Volumetric strain** is a change in volume while distortional strain changes shape. Finite strain may be described as shortening, extension, or shear and may be homogeneous or heterogeneous.

Part C
Essays in 400 words
Answer any 1 questions
Each question carries 12 marks

14. Define foliation and discuss the different types of foliations developed in deformed rocks, explaining their origin and characteristic features.

Foliation is a planar fabric in a rock produced by the preferred orientation of minerals, compositional layering, or closely spaced planar discontinuities. It commonly develops during metamorphism and deformation under directed stress. Foliation may be penetrative at hand-specimen or microscopic scale and is important for interpreting metamorphic and tectonic history.

1. Slaty cleavage: A very fine-grained, closely spaced planar fabric that allows slate to split into

thin sheets. It develops mainly through preferred alignment of tiny platy minerals and pressure-solution processes during low-grade metamorphism. Individual minerals may not be visible to the unaided eye.

2. Phyllitic foliation: Characteristic of phyllite, with a silky sheen and wavy surfaces. It forms as fine mica and chlorite grains grow or become aligned during low- to moderate-grade metamorphism. The minerals are larger than in slate but generally too fine to be readily distinguished individually.

3. Schistosity: A well-developed foliation caused by the parallel alignment of visible platy or elongate minerals,commonly mica, chlorite, or amphibole. It is typical of schist and commonly develops at medium metamorphic grades through mineral growth, rotation, and deformation. The rock splits along mineral-rich planes.

4. Gneissic banding: A compositional foliation consisting of alternating light-coloured quartz- and feldspar-rich bands and darker bands rich in biotite, amphibole, or other mafic minerals. It develops through mineral segregation,metamorphic differentiation, transposition of earlier layering, or high-grade deformation. Unlike schistosity, it is commonly defined by compositional bands rather than only by aligned platy minerals.

5. Crenulation cleavage: A new cleavage formed when an earlier foliation is folded into small-scale crenulations during a later deformation. The new cleavage commonly develops approximately parallel to the axial planes of the microfolds. It is evidence of polyphase deformation.

6. Mylonitic foliation: A planar fabric in strongly sheared rocks formed by intense ductile deformation, grain-size reduction, recrystallisation, and mineral alignment. It is commonly associated with shear zones and may occur with stretching lineation and compositional banding.

Origin and significance: Foliation can form by mineral alignment, rotation and recrystallisation, flattening, pressure solution, metamorphic differentiation, or ductile shearing. Its orientation relative to bedding, fold axial planes,lineation, and shear-sense indicators helps reconstruct deformation events. Not every planar feature has the same origin; primary bedding and tectonic foliation should be distinguished using field relationships and petrography.

15.Summarize the major types of folds recognized in Donath and Parker’s genetic classification, with suitable diagrams and examples.

Donath and Parker’s genetic approach classifies folds according to the deformation mechanism and the way the folded layers accommodate strain, rather than relying only on fold shape.

1.Flexural-slip folds: Layers bend while adjacent beds slide past one another along bedding surfaces. Layer thickness is broadly maintained and bedding surfaces act as slip surfaces. They are common in competent, layered sedimentary successions and may show slickensides or

small offsets along bedding.2.Flexural-flow folds: Bending is accommodated by distributed internal deformation or flow within the layers,rather than only by slip along bedding. This may occur where beds are less competent or where deformation is distributed. Layer thickness may be approximately preserved locally, but internal strain is more important than in a purely flexural -slip fold.3.Passive-flow folds: Layers or markers behave as passive features within a deforming medium and are carried by the overall flow or ductile strain. Fold shape depends on the strain field and the rheology of the material; layering need not control the deformation mechanism. Such folds may occur in strongly deformed metamorphic rocks and ductile shear zones.4. Buckling folds: Layer-parallel shortening causes a competent layer embedded in a weaker matrix to become unstable and buckle. Fold wavelength and amplitude depend on layer thickness, competence contrast, and material properties. Buckling is a common mechanism in multilayered rocks subjected to compression.

Examples and interpretation: Flexural-slip folds are often illustrated by folded sedimentary strata with bedding-plane slip; flexural-flow and passive-flow folds are commonly used to explain ductile deformation in layered or metamorphic rocks; buckling folds develop in competent layers under layer-parallel shortening.